

Neuromorphic FPGA & ASIC Architectures

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Agenda

1. Neuromorphic Accelerator Design Space
2. FPGA Accelerator Overview
3. ASIC ACcelerator Overview

Neuromorphic Accelerator Design Space

Discretization Method

- The common Leaky-Integrate and Fire (LIF) model is defined as an ODE.

$$\tau_m \frac{dV}{dt} = -V(t) + RI(t)$$

- Each hardware designer has to pick a discretization to implement in their hardware.

Neuron Subthreshold Membrane
Dynamics (Gerstner et al. 2014)

Discretization Method

$$\tau_m \frac{dV}{dt} = -V(t) + RI(t)$$


Forward Euler

$$V(t+1) = \left(1 - \frac{\Delta t}{\tau_m}\right) V(t) + \frac{\Delta t}{\tau_m} RI(t)$$

- First-order accurate (error $O(\Delta t^2)$)
- Over-estimates decay as exponential approximated with a straight line.
- Error compounds across timesteps.

Exact Delay / Zero Order Hold (ZOH)

$$V(t+1) = \exp^{-\Delta t/\tau_m} V(t) + (1 - \exp^{-\Delta t/\tau_m}) RI(t)$$

Time-Stepped vs Event-Driven

- Time-Stepped: Update state on every timestep.
- Time-Stepped Leaky-Integrate & Fire Neuron
 - Membrane Update $V(t) = \beta V(t-1) + I(t)$
 - Spike Condition $S(t) = \theta(V(t) - V_{thr})$
 - Membrane Reset $V(t) \leftarrow V(t)(1 - S(t))$
- Event-Driven: Update on spike arrival.
- Event-Driven Leaky Integrate & Fire Neuron
 - Membrane Update:
$$V(t) = V(t_{last}) \exp^{-(t-t_{last})/\tau_m} + I(t)$$
 - Reset condition

Many-core vs Streaming

Memory Architecture

FPGA Neuromorphic Accelerators

Spiker+

YANA

FENN

ASIC Neuromorphic Accelerators

Work #1

Work #2

Work #3